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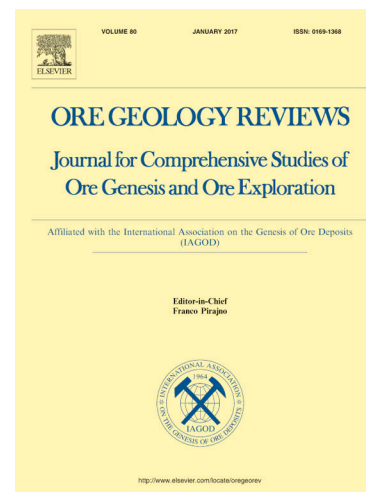
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Genesis of the Maweishan Pb–Zn deposit, eastern China and controls on the distribution and formation of sphalerite-hosted critical metal (Cd, Ga, and In) mineralization

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Abstract

Skarn-type Pb (lead)–Zn (zinc) polymetallic deposits are not only important sources of Pb–Zn but also critical metals such as Cd, In, Ge, and Ga. Significant numbers of skarn-type (or magmato-hydrothermal) Pb–Zn deposits have been identified within the Xuancheng ore district of the Middle and Lower Yangtze River metallogenic belt of eastern China. However, these Pb–Zn deposits are understudied compared to the well-documented porphyry-type Cu–Au, skarn-type Cu–W and Cu–Mo deposits within this district. In addition, the controls and nature of the deportment of critical metals within Pb–Zn deposits also remain poorly constrained, including identification of substitution processes and the sources of these metals within these Pb–Zn systems. The potential genetic link between the Maweishan deposit and the adjacent Qiaomaishan deposit also remains uncertain, with some research considering that the Maweishan deposit is a distal part of the Qiaomaishan skarn system whereas other research suggesting that these are independent skarn systems. This study addresses these issues using new evidence of the timing of mineralization obtained by the in-situ U–Pb dating of the garnet from the Maweishan deposit. This garnet yields a well-defined lower-intercept U–Pb age of 132.3 ± 1.4 Ma, consistent with a zircon U–Pb age for the proximal Maweishan quartz diorite (132.4 ± 0.9 Ma). This combined with the spatial relationships of alteration and mineralization within the deposit strongly suggests that the proximal quartz diorite is genetically related to the formation of the Maweishan deposit and this deposit is not a distal manifestation of the Qiaomaishan skarn system. Sphalerite from the Maweishan deposit also contains high levels of the critical metals Cd [3480–5107 parts per million (ppm)], In (78.3–407 ppm) and Ga (22.4–348 ppm). The Cd²⁺ within this sphalerite was incorporated into the mineral lattice by direct substitution for Zn²⁺, whereas the uniform concentrations of In³⁺ in all three types of sphalerites is indicative of incorporation by the adsorption or capture of homogeneous In nanoparticles. The Ga³⁺ within the Maweishan deposit was mainly incorporated into sphalerite as a result

of coupled substitution with Cu^+ replacing 2Zn^{2+} , with $\text{Ag}^+ + \text{Ga}^{3+}$ coupled substitution for 2Zn^{2+} also involved in the incorporation of Ga into the Sp2 sphalerite. The narrow range of sulfur isotopes ($\delta^{34}\text{S}$ from 4.9‰ to 8.3‰, average of 5.8‰) for sulfides within the Maweishan deposit combined with the new garnet trace element data and elemental maps presented in this study suggests that the metals and fluids that formed the Maweishan were primarily derived primarily from the proximal quartz diorite. These sulfur isotopic data also indicate that a small amount of metals or sulfur were derived from the surrounding country rocks. These data provide new insights into critical metal mineralization associated with Pb–Zn deposits in the Xuancheng ore district, where regional folding concentrated ore-forming fluids that generated both 139 Ma Cu–W and 132 Ma Pb–Zn mineralization events, indicating that this structural zone is an ideal area for future exploration within the Xuancheng district and in other areas with similar characteristics.

Keywords: critical metals; sphalerite; skarn deposit; garnet U–Pb dating; Xuancheng ore district.

1. Introduction

Skarn-type Pb (lead)–Zn (zinc) deposits are one of the important sources of Pb–Zn metal globally (Meinert, 1992; Meinert et al., 2005; Mudd et al., 2017). These deposits are potential sources of numerous critical elements (e.g., Ga, Cd, In), the majority of which are associated with sphalerite (Kelly et al., 2004; Cook et al., 2009; Lockington et al., 2014; Frenzel et al., 2016; Werner et al., 2017a, b; Liu et al., 2022, 2024). The Middle and Lower Yangtze River metallogenic belt (MLYB) is an important belt of mineralization within eastern China and contains numerous porphyry- and skarn-type Cu–(S)–Fe–(Au) deposits (Fig. 1; Mao et al., 2011; Xie et al. 2015; Zhou et al., 2017, 2020; Chang et al., 2019). However, in addition to the Cu–Au–W–Mo deposits, the Xuancheng ore district also hosts numerous intrusion-related Pb–Zn deposits (Fig. 2; Wang et al., 2014; Tang et al., 2022; Li et al., 2024). Previous research in this area has focused on the porphyry-type Chating Cu–Au (Xu et al., 2018, 2022; Xiao et al., 2019, 2021), skarn-type Qiaomaishan Cu–W (Li et al., 2019, 2021a, 2023, 2024; Qi et al., 2020a, b), and Magushan Cu–Mo (Li et al., 2020, 2021b) deposits, meaning the Pb–Zn mineralization in this area remains understudied, including the potential for these deposits to produce significant amounts of critical metals.

The Maweishan Pb–Zn deposit is a representative skarn-type Pb–Zn deposit within the Xuancheng region and contains 64,000 t of contained Zn, 20,300 t of contained Pb, 5,000 t of contained Cu, 80 t of contained Ag, 5.7 million tons of pyrite, and 0.32 million tons of kaolin. However, the genesis of the deposit remains controversial. Although previous research suggests that the Maweishan deposit is a distal part of the Qiaomaishan skarn system as a result of the proximity of the deposit to the Qiaomaishan Cu–W deposit (Fig. 3; Wang et al., 2014; Bao, 2017; Zhou et al., 2019; Bao et al., 2023), a new zircon U–Pb age for a proximal intrusion spatially associated with the Maweishan deposit [132.4 ± 0.9 Ma, mean square weighted deviation (MSWD) = 1.4; Li et al., 2020] contrasts with the age of the Qiaomaishan deposit (139.5 ± 1.2 Ma, MSWD = 0.3; Li et al., 2019). In addition, the in-situ U–Pb ages of scheelite (138.6 ± 3.2 Ma), apatite (139.6 ± 3.3 Ma), and garnet (138.4 ± 1.0 Ma) from the Qiaomaishan deposit has documented the absolute timing of hydrothermal activity and its related mineralization (Li et al.,

2019, 2023). This supports a genetic link between the porphyritic granodiorite proximal to the Qiaomaishan deposit and the formation of the Qiaomaishan deposit itself (Li et al., 2023). However, the zircon U–Pb age of the proximal quartz diorite intrusion associated with the Maweishan deposit represents the timing of the magma emplacement in this area and not necessarily the timing of mineralization. Identification of which intrusion in this area is genetically associated with the formation of the Maweishan deposit by constraining the timing of mineralization is not only important for understanding the genesis of the deposit, but also crucial for exploration for similar deposits (and related critical metal mineralization) in the region and the wider MLYB.

This study aims to examine the long-standing uncertainty about the relationship between the Maweishan and Qiaomaishan deposits and to constrain the deportment, source and formation of the critical metals (Cd, In, and Ga) within sphalerite in the Maweishan deposit. Here we present new in-situ U–Pb dating and geochemical data for garnet, trace element data for sphalerite, and in-situ sulfur isotope data for sulfides from the Maweishan deposit. These new data allow the characterization of the precise timing of mineralization, to determine the existence of any genetic relationship between the Qiaomaishan and Maweishan deposits, and the identification and constraining of mineralizing processes and the source of critical metals within the Maweishan mineralizing system. This study also provides insights into the sources and enrichment process of critical metals associated with skarn-type Pb–Zn systems within the Xuancheng area, the wider MLYB, and beyond.

2. Geological setting

2.1 MLYB and Xuancheng ore district

The MLYB is a major prominent Cu–Fe–Au metallogenic belt in eastern China and the deposits within this belt are typically associated with voluminous Mesozoic magmatic–volcanic activity (Fig. 1; Mao et al., 2011; Xie et al. 2012, 2015; Zhou et al., 2012, 2017; Yang et al., 2014; Pirajno and Zhou, 2015). The deposits within the MLYB are predominantly located in fault-uplifted (e.g. Jiurui, Tongling) and fault-depression (e.g., Ningwu, Luzong) zones, with the fault-uplift zones associated with porphyry–skarn-type Cu–Au deposits and the fault-depression zones associated with porphyry–skarn-type and skarn-type Fe deposits and iron oxide-apatite (IOA) deposits (Fig. 1; Zhou et al., 2017). Previous research (Zhou et al., 2017; Mao et al., 2011) identified three stages of magmatism and its related mineralization within the MLYB: (1) 148–135 Ma granitoids-related Cu–Au–Fe deposits; (2) 135–127 Ma granitoids associated Fe and IOA deposits; and (3) 127–123 Ma A-type granitoids, which are accompanied by minor amounts of U–Au mineralization (Zhou et al., 2017; Li et al., 2024). Porphyry–skarn deposits are a globally important source of critical metals (Werner et al., 2017; Chitalin et al., 2022; McNulty et al., 2022; Schirra et al., 2022; Jowitt, 2024), and the deposits within the MLYB are known to host significant amounts of critical metal mineralization (Fig. 1; Xie et al., 2019, 2024; Yang et al., 2020; Zhou et al., 2020; Guo et al., 2022).

The Xuancheng ore district is located within the southeastern side of the MLYB (Fig. 1; Zhou et al., 2017) and hosts the porphyry-type Chating Cu–Au deposit and the skarn-type Qiaomaishan Cu–W, Maweishan Pb–Zn, Magushan Cu–Mo, Changshan Pb–Zn–Au deposits (Fig. 2). The district is predominantly underlain by Quaternary sediments, with subordinate exposures of early Silurian to

middle Triassic sedimentary units (Fig. 2; Liu and Duan, 2015; Xu et al., 2019). Most of the deposits within the Xuancheng ore district are located within a large-scale anticlinal folding zone with the exception of the Chating deposit (Fig. 2). The intrusions and mineral deposits within the Xuancheng area are structurally controlled by this folding and a series of SE-dipping secondary faults generated by the folding event (Fig. 2; Wang et al., 2019, 2021). The 140–130 Ma intrusions within the Xuancheng ore district are genetically related to the porphyry–skarn-type Cu–Au–W–Mo–Pb–Zn deposits in this region (Fig. 2; Liu and Duan, 2015; Jiang et al., 2017; Li et al., 2019, 2020, 2021a, b, 2023, 2024; Xu et al., 2019; Xiao et al., 2019; Zhou et al., 2019; Qi et al., 2020a, b).

2.2 The Maweishan deposit

The Maweishan Pb–Zn deposit is located adjacent to the Qiaomaishan deposit within the Xuancheng ore district (Figs. 2, 3). The area contains Quaternary sediments and units of the early Carboniferous Gaolishan formation (argillaceous siltstone), the middle Carboniferous Huanglong–Chuanshan formation (limestone containing minor dolomitic limestones), the middle Permian Qixia formation (limestone), and the late Permian Longtan formation (mudstone, siltstone), as identified by geological mapping and exploration drilling (Fig. 3). The deposit is located in a zone of large-scale anticlinal folding within this part of the Xuancheng district, with the core of the fold associated with the deposit containing units of the early Carboniferous Gaolishan formation. The intrusion in this area was emplaced along the lower limb of the fold whereas the mineralization is located within the middle Carboniferous Huanglong–Chuanshan formation in the hinge zone and the upper limb of the fold (Fig. 3C). The study area only contains a few small-scale faults that cross-cut the intrusion (Fig. 3A). Almost all of the economic mineralization within the deposit is hosted by middle Carboniferous limestone units (Fig. 3B, C). This mineralization consists of polymetallic Pb–Zn–Cu–S–Fe sulfide (including sphalerite, galena, pyrite, chalcopyrite, pyrrotite, and arsenopyrite) mineralization and magnetite (Figs. 4, 5) in a gangue mineral assemblage that contains garnet, epidote, actinolite, quartz, calcite, serpentine, and diopside (Figs. 4, 5). Field observations, petrographic characterization, and changes in mineral assemblages allowed the identification of four distinct stages of mineralization (prograde skarn, retrograde skarn, quartz-sulfide, and calcite-sulfide stages) within the Maweishan deposit (Fig. 6).

The intrusion proximal to the Maweishan deposit is a quartz diorite (Fig. 4A). Li et al. (2020) documented that this quartz diorite intrusion was emplaced at 132.4 ± 0.9 Ma using zircon U–Pb dating, with the magmas that formed the intrusion derived from an enriched region of the lithospheric mantle that underwent fractional crystallization and differentiation (Li et al., 2020). The prograde alteration within this area is associated with the formation of garnet and diopside, with the majority of the diopside associated with later retrograde minerals (Fig. 4D). Garnet is mainly found within the exoskarn zone (Fig. 4H). Retrograde hydrous minerals include chlorite, epidote, actinolite, serpentine, and sericite, all of which formed after garnet–diopside skarn alteration, generating widely distributed retrograde hydrous alteration within both endoskarn and exoskarn zones (Figs. 4, 5). The quartz diorite intrusion near the contact zone with the mineralization is also strongly altered to chlorite, sericite, and pyrite (Fig. 4B, C). In addition, the Maweishan deposit endoskarn zone contains some diopside–epidote–quartz assemblages (Fig. 4D–G) that also contain small amounts of sulfides (Fig. 4E). In addition to epidote, garnet skarn and actinolite skarn are also developed within the exoskarn zone (Fig. 4H–L). Epidote, actinolite, and

serpentine are also widespread throughout the exoskarn alteration zones, and epidote–magnetite, actinolite–pyrite, and serpentine–magnetite–pyrite assemblages are also common (Figs. 4I–L; 5A–C).

The formation of the mineralization within the Maweishan deposit is split into quartz–sulfide and calcite–sulfide stages (Figs. 6). Although chalcopyrite is associated with the pyrite–pyrrhotite±quartz assemblages that formed during the early period of the Maweishan hydrothermal activity (Fig. 5D–F), the low amount of Cu present within the Maweishan deposit means that the sphalerite and galena in the deposit have significantly higher economic value than the chalcopyrite. The sulfide mineralization stage is associated with the formation of sphalerite–pyrite–pyrrhotite–galena assemblages and massive or vein-type sulfide ores that formed from this stage until the end of the mineralizing period (Fig. 5F–I). In the Maweishan deposit, there is no obvious difference in sulfide assemblages except that there was more pyrrhotite formed during the quartz–sulfide stage than the calcite–sulfide stage and minor amounts of arsenopyrite formed during the later part of the calcite–sulfide stage of mineralization but at no other time during the formation of the deposit. This means that the main distinction between quartz–sulfide and calcite–sulfide mineralization stages is related to changes in gangue mineral assemblages.

3. Samples and results

3.1 Samples

The petrographic characteristics of garnet within the Maweishan deposit allow the identification of two main types of garnet in the exoskarn zone (Fig. 7). Brown garnet (Grt 1) is hosted by massive garnet skarn with very minor amounts of sulfide visible in hand specimen (Fig. 4G). In contrast, the dark green garnet (Grt 2) is spatially related to sulfide mineralization (Fig. 4H). The brown garnet is often spatially associated with epidote and quartz, and the gaps within this garnet are often filled with epidote and quartz (Fig. 7A–F). In contrast, the Grt 2 garnet is spatially related to magnetite and pyrite (Fig. 7G–I). The Grt 1 garnet appears brown in hand specimen (Fig. 4G), colorless under plane-polarized light, and has oscillatory zoning visible under cross-polarized light (Fig. 7A, B, D, E). This Grt 1 garnet also has generally homogenous compositions as evidenced by their uniform appearance during backscattered electron (BSE) imaging (Fig. 7C, F). The dark green Grt 2 garnet (Fig. 4H) appears yellow under plane-polarized light, and is completely isotropic and is free of any visible any oscillatory zoning under cross-polarized light (Fig. 7G, H). The BSE imaging of this garnet also indicates that the Grt 2 garnet is chemically homogeneous and is free of any oscillatory zoned structure defined by geochemical variations (Fig. 7I).

Similar to garnet, the sphalerite in Maweishan deposit also has different petrographic characteristics and associations with different mineral assemblages. The color of the sphalerite and variations in gangue mineral associations also allow classification of different types of sphalerite, with the quartz–sulfide stage sphalerite (Sp 1) being dark red, opaque and/or semitransparent under plane polarized light and is associated with a sphalerite–pyrite–galena–quartz assemblage (Fig. 8A–C). The deposit also hosts a transitional sphalerite (hereafter Sp 2) that is similar in color to Sp 1 but is associated with a gangue mineral assemblage that contains calcite instead of quartz (Fig. 8D–F). The final calcite–sulfide stage sphalerite (Sp 3) appears yellow-brown and translucent under plane polarized light and is associated with calcite (Fig. 8G–I) and a small amount of arsenopyrite.

In this study, the two generations of garnet identified within the Maweishan deposit were selected for geochemistry compositional analysis, elemental mapping, and in-situ U–Pb dating. In addition, the trace element concentrations of sphalerite and the S isotopic compositions of a range of sulfides were determined using in-situ analysis. Details of the analytical methods and parameters used are given in Table S1 and Text S1 in Supplemental Material 1 (SM 1) and the results of these analyses are given in Table S2 to S6 in SM 1, along with the results of analysis of reference standards that provide information on the precision and accuracy of these analyses. The results of these analyses are described below.

3.1 Garnet petrographic and geochemical characteristics

The majority of the garnets within the Maweishan deposit compositionally vary between grossularite ($\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) and andradite ($\text{Ca}_3\text{Fe}_2\text{Si}_3\text{O}_{12}$) end-members (Fig. 9A; Table S2 in SM 1). These two generations of garnet also have significant differences in major element composition. The Grt 1 exoskarn garnets contain 8.29–13.38 wt.% Al_2O_3 (N = 16) and has $\text{Adr}_{42-64}\text{Grs}_{36-58}$ compositions (Fig. 9A) whereas the Grt 2 exoskarn garnets contain relatively low concentrations of Al_2O_3 (0–0.28 wt.%, N = 14) and have almost pure andradite compositions (Fig. 9A).

The Grt 1 and Grt 2 garnets also have different rare earth element (REE) concentrations and patterns (Fig. 9B; normalized to the C1 chondrite composition of McDonough and Sun, 1995). Grt 1 is light REE (LREE) enriched but is heavy REE (HREE) depleted with weakly positive Eu anomalies (0.48 to 1.67, average at 1.02; Eu anomalies = $\frac{\text{Eu}_{\text{sample}}/\text{Eu}_{\text{chondrite}}}{\sqrt{(\text{Sm}_{\text{sample}}/\text{Sm}_{\text{chondrite}}) \times (\text{Gd}_{\text{sample}}/\text{Gd}_{\text{chondrite}})}}$; N = 20). The Grt 1 garnet also has chondrite-normalized REE patterns with obvious peaks at Pr and Nd (Fig. 9B). In comparison, the Grt 2 garnet contains very low concentrations of the REE and basically no HREE (consistently below the detection limit; Table S2 in SM 1). The concentrations of U and W within the Grt 1 garnet are also lower than the concentrations of these elements in the Grt 2 garnet, but the Sc, V, Mn and Zr concentrations within the Grt 1 garnet are higher than in the Grt 2 garnet, and the concentrations of Co, Ni, Cu, Zn and Mo are low (mostly below the detection limit) in both Grt 1 and Grt 2 garnet (Fig. 9C; Table S3 in SM 1).

The BSE imaging undertaken during this study also identified structures where grossularite garnet (Grt 1) was encased by andradite garnet (Grt 2; Fig. 10A, B). This elemental mapping also indicates that the grossularite garnet has higher Al, Ti, REE, and Y concentrations than the andradite garnet (Grt 2; Fig. 10C–G). This imaging also identified elevated concentrations of Cu and Zn within fractures and veins in the garnet (Fig. 10H, I).

3.2 Garnet in-situ U–Pb dating

The U–Pb ages for the garnets analyzed during this study were calculated using a linear regression within Tera–Wasserburg concordia diagrams (Tera and Wasserburg, 1972). The U–Pb isotopic data for the garnet from the Maweishan deposit yields a well-defined lower-intercept age of 132.3 ± 1.4 Ma (2σ , N=37, MSWD = 1.8; Fig. 11; Table S4 in SM 1), and the data for the reference standards used during

this study are also provided in Table S4 in SM 1.

3.3 Geochemistry of sphalerite

The concentrations of some elements such as Si, Ti, Co, Ni, Mo, and W that are present in very low abundances in the sphalerite in the study area (i.e., are below the detection limit) are not given in Table S5 in SM 1. The concentrations of Ge, As, and Sb in the three types of sphalerite within the Maweishan deposit are also very low (Table S5 in SM 1), but the concentrations of Cd, Ga, In, Cu, Mn and Ag (except Zn and Fe) are relatively high (Table S5 in SM 1). The Sp 1 sphalerite (N = 12) that formed during the quartz–sulfide stage has the highest concentrations of Fe (6.9 to 7.3 wt.%, data from LA–ICP–MS; Fig. 12A), Mn (3639–4256 ppm; Fig. 12B) and Cd (4172–55107 ppm; Fig. 12C), but the lowest concentrations of Ag (6.11–23.9 ppm, Fig. 12D) and Ga (22.4–164 ppm; Fig. 12E) of the sphalerite analyzed during this study. The Sp 2 sphalerite (N = 14) has moderate concentrations of Fe (4.5 to 7.1 wt.%; Fig. 12A), Mn (2246–5191 ppm; Fig. 12B), and Ga (103–230 ppm; Fig. 12E), high concentrations of Ag (67.5–187 ppm; Fig. 12D) and low concentrations of In (101–142 ppm; Fig. 12F). The concentration of Cd (3642–4211 ppm; Fig. 12C) within the Sp 2 sphalerite is slightly lower than in the Sp 1 sphalerite. The Sp 3 sphalerite (N = 29) has the lowest concentrations of Fe (3.2 to 4.5 wt.%; Fig. 12A) of any of the sphalerite analyzed during this study although this sphalerite has Ga (54.6–348 ppm; Fig. 12E) and Cd (3480–4333 ppm; Fig. 12C) concentrations similar to those within the Sp 2 sphalerite.

3.4 In-situ sulfur isotopes

Sulfide $\delta^{34}\text{S}$ values from samples from the quartz–sulfide and calcite–sulfide stages of formation of the Maweishan deposit range from 4.9‰ to 8.3‰ with distinct ranges for pyrrhotite (5.8‰ to 6.1‰; N = 4, average = 6.0‰), chalcopyrite (5.3‰ to 8.1‰; N = 5, average = 7.0‰), pyrite (4.9‰ to 8.3‰; N = 6, average = 6.9‰), and sphalerite (5.2‰–6.3‰, N = 5 with an average of 5.8‰; Fig. 13; Table S6 in SM 1).

4. Discussion

4.1 Timing of mineralization within the Maweishan deposit and relationships between the proximal quartz diorite and the Qiaomaishan skarn system

As mentioned above, the spatial proximity between the Maweishan and Qiaomaishan deposits (Figs. 2, 3) has led to some researchers suggesting that these deposits represent distal and proximal components of the same skarn hydrothermal system, respectively. This model would also suggest that both of these deposits are genetically related to the Qiaomaishan porphyritic granodiorite (Wang et al., 2014; Bao, 2017; Zhou et al., 2019; Bao et al., 2023). Although Li et al. (2020) documented the geochemistry and zircon U–Pb ages of the quartz diorite intrusion spatially proximal to the Maweishan deposit, no direct dating of the skarn alteration or mineralization within the deposit has been undertaken before this study. This lack of geochronological data for correlation purposes means that the status of the Maweishan

deposit, the presence of a genetic relationship between the deposit and the quartz diorite, and the relationship (if any) with the Qiaomaishan deposit remains uncertain. The analysis of garnet during this study yielded a well-defined lower intercept U–Pb age of 132.3 ± 1.4 Ma (Fig. 11) for the Maweishan hydrothermal event that formed the deposit, which is within analytical uncertainty of the zircon U–Pb age of 132.4 ± 0.9 Ma (MSWD = 1.4, N = 29; Li et al., 2020) for the proximal quartz diorite intrusion. However, this age differs significantly from the emplacement age of the Qiaomaishan porphyritic granodiorite (139.5 ± 1.2 Ma; Li et al., 2019), suggesting that the quartz diorite intrusion proximal to the Maweishan deposit is genetically related to the deposit (including the formation of the garnet skarn). The paragenetic assemblages and cross-cutting relationships documented within the garnet skarn and sulfide mineralization (Fig. 5D, E) in the Maweishan deposit also provides evidence of the intimate relationships between these two styles of alteration. All of this suggests that the alteration and mineralization within the Maweishan deposit is closely genetically related to the proximal quartz diorite intrusion. The timing of mineralization within the Maweishan deposit also suggests that the Maweishan and Qiaomaishan deposits represent two distinctly separate magmatic–hydrothermal–mineralization events and systems.

4.2 Deportment and substitution mechanisms of Cd, In, and Ga elements

Previous research has identified some of the controls on the deportment and substitution processes involved in the incorporation of elements, including economically important critical metals, into the sphalerite crystal structure (Di Benedetto et al., 2005; Cook et al., 2009; Pfaff et al., 2011; Murakami and Ishihara, 2013; Mudd et al., 2017; Werner et al., 2017a, b; Bauer et al., 2019; Filimonova et al., 2019; Yu et al., 2023). The trace elements within sphalerite primarily exist in three forms: (1) lattice-bound solid solutions (isomorphic substitution), (2) nano-scale clusters, or (3) micro-mineral inclusions (Cook et al., 2009, 2016; Lockington et al., 2014; George et al., 2016; Liu et al., 2022; Wang et al., 2024). Laser ablation profiles can provide diagnostic criteria to distinguish these modes of incorporation where homogeneous element distribution with smooth signals parallel to Zn is typically indicative of isomorphic substitution or uniformly dispersed nanoparticles (Cook et al., 2009; Ye et al., 2011; Liu et al., 2022; Wang et al., 2024), whereas anomalous signal spikes discordant with Zn are diagnostic of the presence of discrete micro-inclusions containing the elements in question (George et al., 2016; Wang et al., 2024). In this study, the Fe, Mn, Cd, In, Ga, Cu, and Ag data for all types of sphalerite within the Maweishan deposit yield relatively flat LA–ICP–MS signal spectra (Fig. 14), suggesting that these elements may have been incorporated into sphalerite either by the incorporation of homogeneously distributed nanoparticles or by isomorphic substitution.

The three types sphalerite within the Maweishan deposit all contain relatively high concentrations of critical metals such as Cd, Ga, and In (Fig. 12C, E, F), but the processes by which these critical elements have been incorporated into sphalerite are not the same. The incorporation of Cd^{2+} into sphalerite is either directly by substitution for Zn^{2+} (Cook et al., 2009; Wright and Gale, 2010; Liu et al., 2022; Wang et al., 2024) or by the complex coupled substitution of $\text{Cd}^{2+} + \text{Fe}^{2+}$ for 2Zn^{2+} (Bonnet et al., 2016). The Cd concentrations within the sphalerite from the Maweishan deposit do not correlate with Fe and Mn (Fig. 15B, C) but the Sp 2 sphalerite yields Cd concentrations that correlate with Zn concentrations, reflecting the wider range of Zn concentrations within this sphalerite that more clearly defines this trend than is the case for the Sp 1 and Sp 3 sphalerite. The presence of this Zn–Cd correlation suggests that the Cd in the

sphalerite within the Maweishan deposit was incorporated by direct substitution for Zn rather than by complex substitution involving Fe (or Mn; Bonnet et al., 2016). The different Cd concentrations in the three types of sphalerite present in the study area also indicates that the substitution process of Cd and Zn is controlled by factors other than simple temperature differences (Fig. 12C). Previous studies (Cook et al., 2009, 2016; Belissont et al., 2014; Lockington et al., 2014; Frenzel et al., 2016, 2020; Wang et al., 2024) documented that Cd concentrations in sphalerite are controlled by changes in conditions such as temperature and fluid properties. The early-formed Sp 1 sphalerite associated with quartz (Fig. 8A–C), suggesting higher temperatures than the Sp 2 and Sp 3 sphalerite associated with calcite (Fig. 8D–I; Kwak, 1986; Pan and Dong, 1999; Meinert et al., 2005). However, although the Sp 1 sphalerite has the highest Cd content of the three types of sphalerite in the study area, the Sp 2 and Sp 3 sphalerite contain similar concentrations of Cd (Fig. 12C). This obviously cannot be explained by changes in temperature conditions, and variations in the abundance of Cd in the ore-forming fluids as a control on the substitution of Cd into sphalerite seems to be a more plausible explanation for these variations. The evolution of the hydrothermal system before the quartz–sulfide stage of mineralization recorded within the Maweishan deposit (Fig. 6) did not involve any significant sulfide (especially sphalerite) precipitation, meaning that Cd was enriched in the residual fluid. This led to the earliest formed sphalerite having the highest Cd content of any of the sphalerite within the study area (Fig. 12C). The precipitation of the Sp 1 sphalerite reduced the Cd concentration in the residual hydrothermal fluid, leading to the later formation of Sp 2 and 3 sphalerite with lower concentrations of Cd, also indicating that the system itself was likely closed (at least relating to Cd) during the formation of these different stages of mineralization (Fig. 12C).

The different valence states of In^{3+} and Ga^{3+} compared to Zn^{2+} means that these elements are usually incorporated into sphalerite by the coupled substitution replacement of 2Zn^{2+} by $(\text{Cu}, \text{Ag})^+ + (\text{In}, \text{Ga})^{3+}$ (Cook et al., 2009; Wang et al., 2024). The positive correlation of $\text{Cu}+\text{Ag}$ and $\text{Ga}+\text{In}$ indicates that the $(\text{Cu}, \text{Ag})^+ + (\text{In}, \text{Ga})^{3+}$ and 2Zn^{2+} substitution process may be the cause of the In and Ga enrichments in sphalerite within the Maweishan deposit (Fig. 15D). The concentrations of $\text{Cu}+\text{Ag}$ also positively correlate with Ga concentrations (Fig. 15E), but the Ga contents of Sp 1 and Sp 2 Ga in the three types of sphalerite lack correlation with Ag (Fig. 15F). This is especially true of the concentrations of Ag in the Sp 1 sphalerite (6.11 to 23.9 ppm; Table S5 in SM 1), which are significantly lower than the concentrations of Ga in the same sphalerite (22.4 to 164 ppm; Table S5 in SM 1). This suggests that the substitution of $(\text{Ag}^+ + \text{Ga}^{3+})$ replacing 2Zn^{2+} was not the dominant process controlling Ga concentrations during the formation of the Sp 1 and Sp 3 sphalerite at least. This contrasts with the fact that the Ag and Ga concentrations within the Sp 2 sphalerite positively correlate (Fig 15F), indicating that the substitution of $(\text{Ag}^+ + \text{Ga}^{3+})$ for 2Zn^{2+} may have occurred during the formation of the Sp 2 sphalerite. There are also strong positive correlations between Cu and Ga concentrations in all three types of sphalerite (barring a few minor discrete data points; Fig. 15G), which suggests that the substitution of $(\text{Cu}^+ + \text{Ga}^{3+})$ for 2Zn^{2+} was also an important process in the incorporation of Ga into sphalerite in the study area. In summary, the main process whereby Ga was incorporated into sphalerite was the replacement of 2Zn^{2+} by $(\text{Cu}^+ + \text{Ga}^{3+})$, a process that dominated the Ga budget of sphalerite within the Maweishan deposit. In addition, Ga was incorporated into Sp 2 sphalerite by the substitution of $(\text{Cu}^+ + \text{Ga}^{3+})$ and $(\text{Ag}^+ + \text{Ga}^{3+})$ for 2Zn^{2+} .

All three types of sphalerite in the Maweishan deposit have essentially uniform concentrations of In (Fig. 12F) that do not correlate with variations in the concentration of Cu, Ag and other elements (Fig. 15H, I; Table S5 in SM 1). This lack of correlation suggests that the In within this sphalerite may not

have been incorporated into sphalerite by substitution but instead is more likely to have been incorporated by adsorption or by the capturing of nanoparticles during sphalerite crystallization. This process means that the concentration of In within the resulting sphalerite is therefore more dependent on the fluid properties that formed the sphalerite rather than any variations in temperature or other factors. The fact that In was not incorporated into sphalerite by substitution like other elements (e.g., Cd, Ga) and thus was not incorporated in significant quantities in the early-formed sphalerite means that the residual hydrothermal fluid within the closed and evolving Maweishan mineralizing system contained higher concentrations of In than during the earlier stages of mineralization, forming discrete areas of the Sp 3 sphalerite that contain relatively high concentrations of In (Fig. 12F).

4.3 Sources of fluid and metals

The sulfides present within the Maweishan deposit, including chalcopyrite, pyrite, pyrrhotite, galena, and sphalerite (Figs. 4, 5), formed during both the quartz–sulfide and calcite–sulfide phases of mineralization and the deposit is free of sulfate. These assemblages are indicative of reduced H_2S -dominated hydrothermal conditions (Ohmoto, 1972; Ohmoto and Goldhaber, 1997), and the $\delta^{34}\text{S}$ composition of the sulfides from the Maweishan deposit can thus reflect the S isotopic characteristics of the hydrothermal fluid. The sulfides within the Maweishan deposit yield a narrow range of $\delta^{34}\text{S}$ values (4.9‰–8.3‰ with a mean of 5.8‰; $N = 20$) that is slightly heavier than those expected for magmatic-derived fluids (–5 to +5‰; Fig. 12; Meinert et al., 2003, 2005; Hoefs, 2009; Cooke et al., 2011; Parsapoor et al., 2014). This suggests that the most of sulfur (and ore-forming metals) within the fluids that formed the Maweishan deposit was derived from a magmatic source, with minor contributions from the surrounding country rock that move the S isotopic composition to slightly heavier ranges than expected for a purely magma-derived mineralizing system. This study also constrains the timing of mineralization of the Maweishan deposit (Fig. 10A) and demonstrates that the proximal quartz diorite intrusion in the study area is genetically related to the Maweishan deposit, suggesting that this intrusion was the source of the fluids that formed the deposit as well as the metals within the deposit.

This model is further supported by the fact that the REE patterns of the exoskarn Grt 1 garnet within the Maweishan deposit are similar to the REE patterns of the genetically associated quartz diorite (Fig. 9B), providing further evidence of the genetic links between the exoskarn garnet and the quartz diorite. This not only indicates that the quartz diorite is genetically related to the formation of the Maweishan deposit, but also suggests a genetic link between the fluids that formed the Maweishan skarn system and the quartz diorite. Although the REE concentrations in the Grt 2 garnet are very low, this garnet contains low concentrations of the LREE and HREE concentrations that are below the detection limit, which is similar with the REE patterns of the quartz diorite and the Grt 1 garnet (Fig. 9B; Table S3 in SM 1). The Grt 1 grossularite garnet is also rimmed or encased by Grt 2 andradite garnet (Fig. 10), which also indicates that the formation of both of the generations of garnet is genetically related to the quartz diorite. In addition, the elemental mapping of garnet from the Maweishan deposit indicates that the fractures and interstitial spaces within the garnet contains high concentrations of Cu and Zn (Fig. 10H, I), suggesting that the ore-forming fluids associated with sulfide mineralization (after garnet formation) did not migrate long distances, but formed sulfide mineralization within the skarn zone, consistent with the at least relatively or partially closed system behavior indicated by the substitution processes outlined above. This

is also consistent with the characteristics observed in the Maweishan deposit, such as the sulfide associated with or cross-cut early actinolite and garnet skarn (Figs. 4H, J–L, 5D, E, 7G–I).

4.4 Implications and insights for further exploration

The uncertainty over whether the Maweishan and Qiaomaishan deposits are part of the same skarn system has been propagated as a result of a lack of decisive evidence and constraints to resolve this issue (Wang et al., 2014; Bao, 2017; Zhou et al., 2019; Bao et al., 2023). This study directly constrains the timing of mineralization and formation of the Maweishan deposit and establishes that the Maweishan and Qiaomaishan deposits represent two separate skarn systems. This indicates that the regional folding that forms a focus for mineralization within the Xuancheng ore district not only involved the ~139 Ma Cu–W mineralizing event but also the ~132 Ma Pb–Zn polymetallic mineralization event identified within this region. The close proximity of the Maweishan and Qiaomaishan skarn systems and the relatively large time span of mineralization (139 and 132 Ma) in this area indicates that the regional folding that focused both mineralizing events was not significantly changed by any large-scale tectonism in this area between the 139 and 132 Ma events, enabling the intrusion of the quartz diorite and the formation of the Maweishan deposit at 132 Ma with similar characteristics to the earlier formed Qiaomaishan deposit. This indicates that the regional folding within the Xuancheng ore district represents a significant area for future exploration, especially along the contact between limestones and early Cretaceous intrusions within this structural focus area.

Previous research has also constrained the timing of Cu–Au, Cu–W, and Cu–Mo mineralizing events within the Xuancheng region (Jiang et al., 2017; Li et al., 2021, 2023), but there is still a lack of research on the Pb–Zn deposits within this area. This research provides the first data of the timing of Pb–Zn mineralization events within the Xuancheng area. Combining the data presented in this study with that generated by previous research, including identifying the mineralization events within the Xuancheng district, highlights the prospectivity of this ore district for further mineral exploration. These data also suggest that the intrusions in this region formed during the early Cretaceous should be considered high priority targets for skarn-type mineral exploration, with early (before 136 Ma) magmatism primarily associated with Cu polymetallic mineralization events whereas late (after 136 Ma) magmatism-related mineralization events are dominated by Pb–Zn mineralization.

The critical element potential of this region also requires further assessment. The three types of sphalerites from the Maweishan deposit all contain relatively high concentrations of Cd, In, and Ga (Fig. 11C, E, F), the majority of which was derived from a magmatic source. This clearly demonstrates the considerable potential for critical metal mineralization within the Xuancheng ore district. This has already been demonstrated by Tang et al. (2022), who determined that the Changshan Pb–Zn deposit in the Xuancheng ore district contains resources with more than 800 t contained Cd. All of this suggests that the critical metal mineralization associated with the Pb–Zn deposits in the Xuancheng ore district should be examined further both during exploration and in additional research.

5. Conclusions

(1) The mineralization within the Maweishan deposit formed at 132.3 ± 1.4 Ma, consistent with the zircon U-Pb age of the proximal quartz diorite intrusion (132.4 ± 0.9 Ma). This indicates that the quartz diorite intrusion is genetically related to the formation of the Maweishan deposit and that Maweishan deposit is not a distal representation of the Qiaomaishan skarn system.

(2) Sphalerite within the Maweishan deposit contains high concentrations of the critical metals Cd, In and Ga. The Cd^{2+} within this sphalerite is incorporated as a result of direct substitution for Zn^{2+} , whereas the uniform concentrations of In^{3+} is indicative of incorporation of In as homogeneous nanoparticles through adsorption or capture during the crystallization of sphalerite. The Ga^{3+} within the Maweishan deposit was mainly incorporated into sphalerite as a result of coupled substitution with Cu^+ replacing 2Zn^{2+} , and $\text{Ag}^+ + \text{Ga}^{3+}$ coupled substitution of 2Zn^{2+} also playing a role during the formation of Sp₂.

(3) The metals and fluids that formed the Maweishan deposit were primarily derived from the proximal quartz diorite intrusion. Sulfur isotopic data also indicate that a small amount of metals or sulfur were derived from the surrounding country rocks.

(4) The regional folding structural zone within the Xuancheng district focused both 139 Ma Cu–W and 132 Ma Pb–Zn mineralizing systems and hydrothermal fluids, indicating that this structural zone represents a key focus area that is highly prospective for future mineral exploration within the Xuancheng district. The critical metal mineralization associated with the numerous Pb–Zn deposits within the Xuancheng district also needs to be further examined to determine the potential of these deposits to be key sources of these crucial commodities.

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Figure captions

Fig. 1. Distribution of critical metal-bearing deposits within the MLYB and the location of the Xuancheng ore district (modified after Mao et al., 2011; Fan et al., 2014; Xie et al., 2019, 2020; Zhou et al., 2020).

Fig. 2. Map showing the geology of the regional folding tectonic belt within Xuancheng ore district (modified after Liu and Duan, 2015).

Fig. 3. Geological map of the area around the Maweishan deposit (A, B) and cross-section of geological transect (C).

Fig. 4. (A) Photograph showing representative examples of the quartz diorite in the study area. (B–C) Photograph and photomicrograph showing representative examples of strong altered quartz diorite. (D–G) Photographs and photomicrographs of representative endoskarn diopside + epidote + quartz + chalcopyrite + pyrite assemblages within the Maweishan deposit. (H) Photograph showing representative examples of garnet exoskarn. (I) Photograph showing epidote exoskarn with magnetite mineralization. (J–L) Photographs and photomicrographs of actinolite endoskarn with pyrite mineralization. (E) was taken under reflected light and (C, K) and (F, L) were taken under transmitted plane-polarized light and transmitted cross-polarized light, respectively. Abbreviations are as follows: Py = pyrite, Ser = sericite, Chl = chlorite, Di = diopside, Ep = epidote, Qtz = quartz, Ccp = chalcopyrite, Grt = garnet, Mt = magnetite, and Act = actinolite.

Fig. 5. (A–C) Photographs and photomicrographs showing serpentine + magnetite + pyrite assemblages. (D, E) Quartz + pyrite + chalcopyrite coarse veins cut through early-formed garnet skarn. (F) Representative massive sulfide ore. (G) Photomicrograph showing the pyrite + pyrrhotite assemblages.

(H, I) Representative massive Pb–Zn sulfide ore. (B, G) were taken under reflected light and (C) was taken under transmitted cross-polarized light, respectively. Abbreviations are in Fig. 4 with Srp = serpentine, Po = pyrrhotite, Sp = sphalerite, and Gn = galena.

Fig. 6. Paragenetic sequence of mineralization and alteration recorded within the Maweishan deposit.

Fig. 7. (A–F) Representative photomicrographs and BSE images of Grt 1 + quartz + epidote assemblages within the Maweishan deposit. (G–I) Photomicrographs and a BSE image showing Grt 2 + magnetite + pyrite assemblages. (A, D, G) and (B, E, H) were taken under transmitted plane-polarized light and transmitted cross-polarized light, respectively. Abbreviations are as in Fig. 4.

Fig. 8. (A–C) Photomicrographs showing the sphalerite + galena + pyrite + quartz assemblages associated with Sp 1 sphalerite. (D–F) Photomicrographs indicating variations in the gangue mineral assemblage between Sp 2 and Sp 1 sphalerite although both forms of sphalerite are similar in color. (G–I) Photomicrographs showing the sphalerite + galena + pyrite + calcite assemblages associated with Sp 3 sphalerite. (A, D, G) were taken under reflected light and (B, E, H) and (C, F, I) were taken under transmitted plane-polarized light and transmitted cross-polarized light, respectively. Abbreviations are as in Figs. 4, 5 with Cal = calcite.

Fig. 9. (A) Ternary diagram showing variations in the composition of garnet within the Maweishan deposit. (B) REE patterns for Grt 1 garnet from the Maweishan deposit normalized to the C1 chondrite composition of McDonough and Sun (1995). These diagrams also show the composition of the Maweishan quartz diorite (Li et al., 2020) and samples of Carboniferous carbonate rocks from the study area (Du et al., 2014). (C) Box diagrams outlining variations in Sc, V, Zr, U, and W concentrations within the garnet within the Maweishan deposit.

Fig. 10. Results of the LA–ICP–MS elemental mapping of the encasing structure of Grt 1 and Grt 2 garnet within the Maweishan deposit. (A) was taken under cross-polarized light. (C–F) Elemental maps indicating the distribution of selected trace elements within this garnet.

Fig. 11. Tera–Wasserburg diagrams of the U–Pb ages for garnet from the Maweishan deposit.

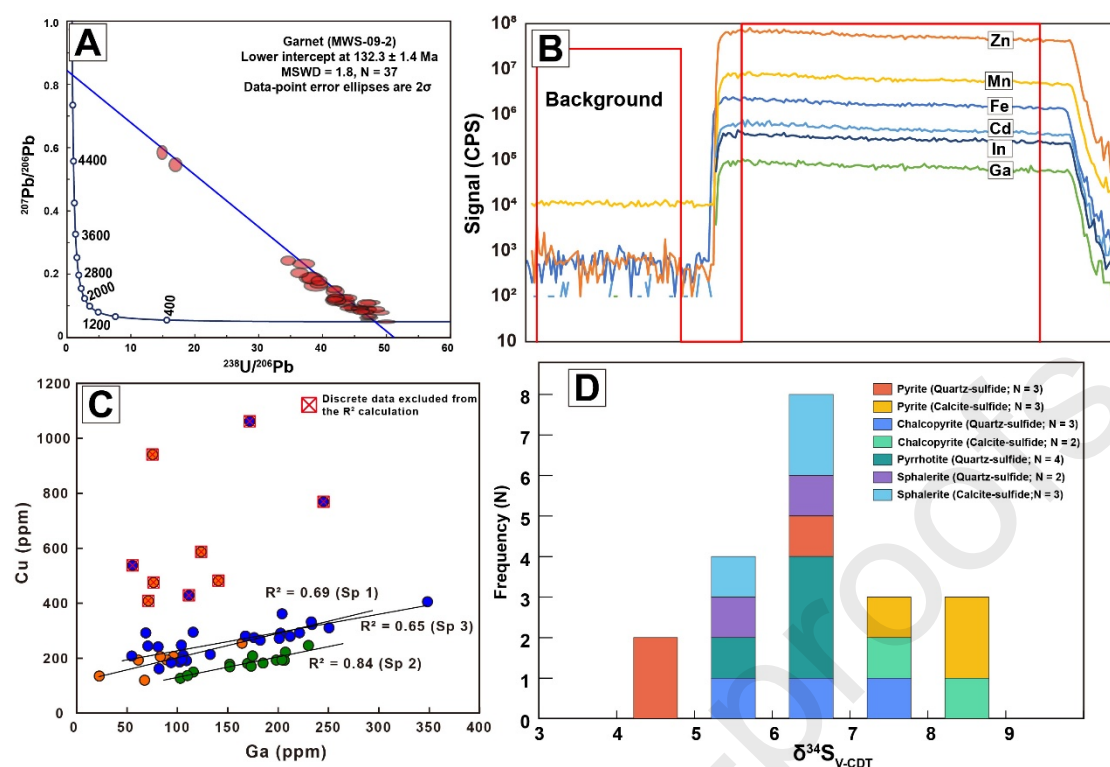
Fig. 12. Box diagrams showing the major and trace element concentrations of three types of sphalerite within the Maweishan deposit.

Fig. 13. Histogram of $\delta^{34}\text{S}$ values of sulfides from the Maweishan deposit.

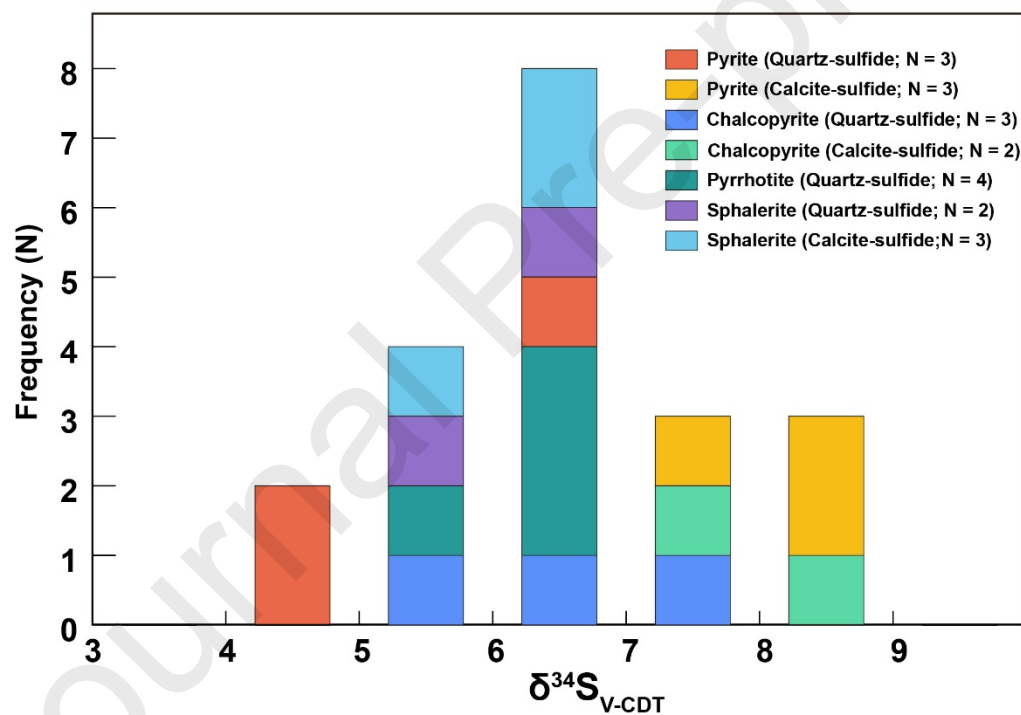
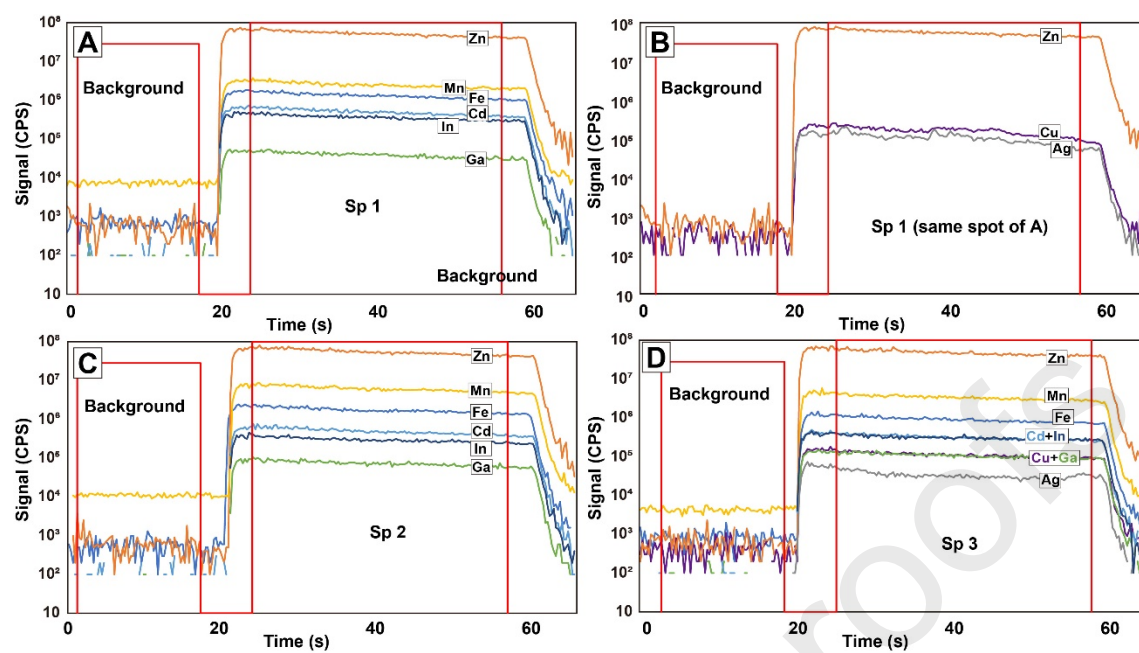
Fig. 14. Typical LA–ICP–MS signal spectra for three types of sphalerite within the Maweishan deposit.

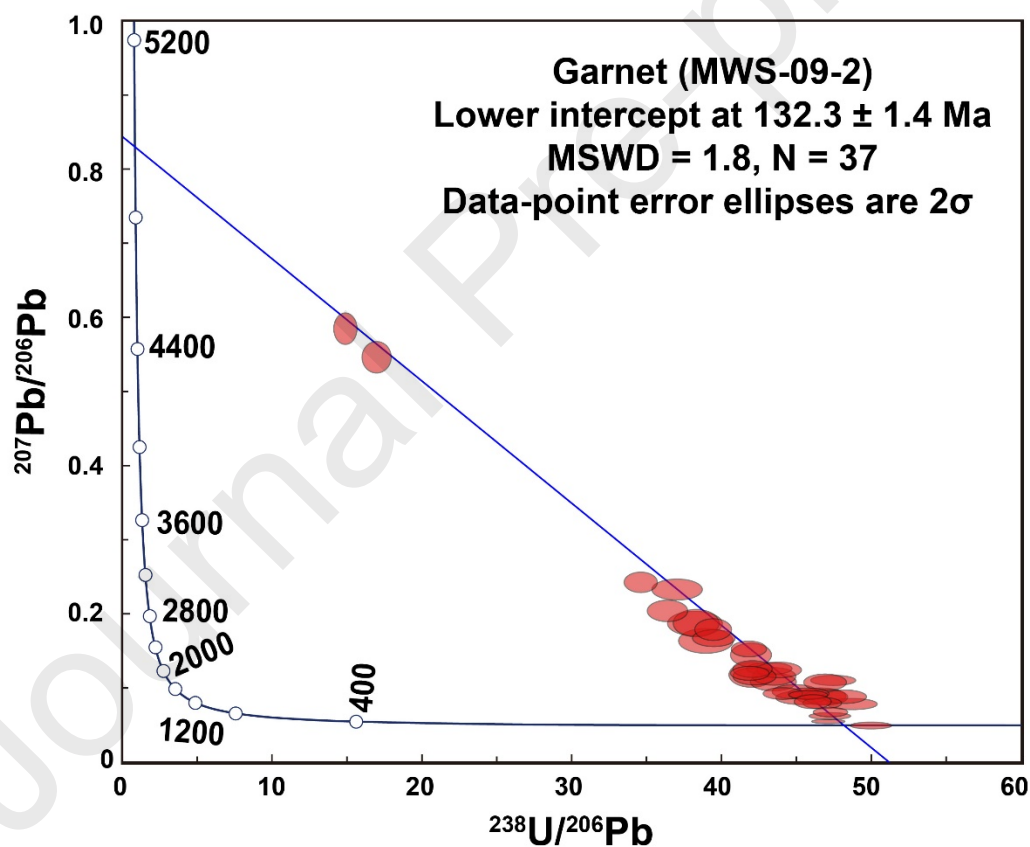
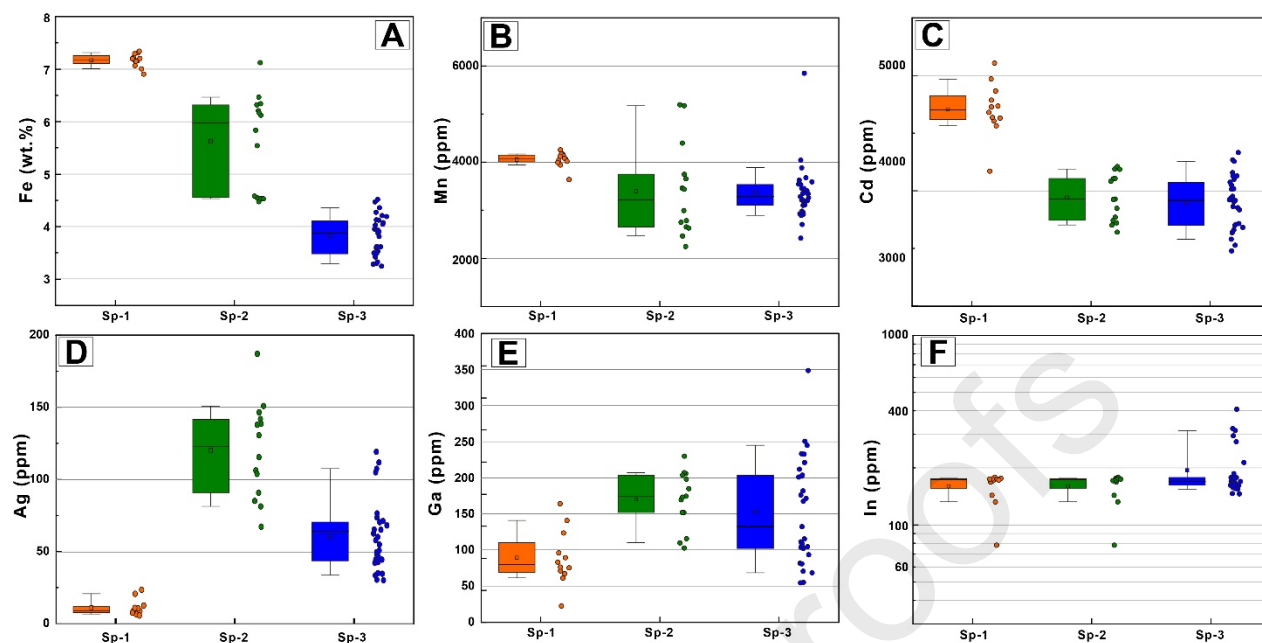
Fig. 15. Binary plots indicating variations in concentrations of elements within sphalerite from the Maweishan deposit. (A) Zn versus (vs.) Cd, (B) Fe vs. Cd, (C) Mn vs. Cd, (D) (Ga+In) vs. (Cu+Ag), (E) Ga vs. (Cu+Ag), (F) Ga vs. Ag, (G) Ga vs. Cu, (H) In vs. Cu, and (I) In vs. Ag.

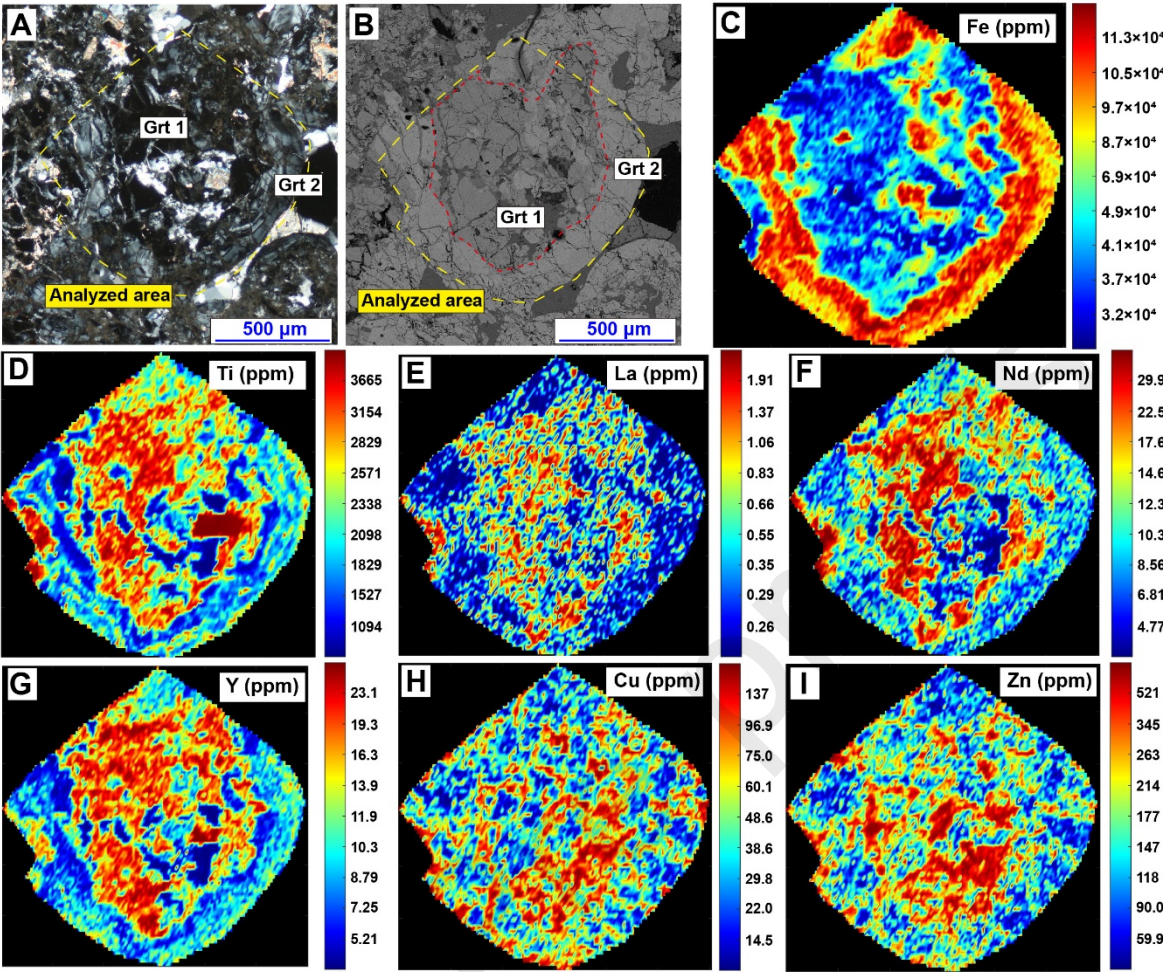
1. The Maweishan deposit formed at 132.3 ± 1.4 Ma, consist with a zircon U–Pb age for the proximal quartz diorite (132.4 ± 0.9 Ma), suggests that the Maweishan deposit is not a distal part of the Qiaomaishan system.
2. Sphalerite within the Maweishan deposit incorporates Cd^{2+} by direct substitution for Zn^{2+} , whereas In^{3+} are incorporated as homogeneous nanoparticles through adsorption or capture.
3. The sphalerite contains Ga^{3+} incorporated as a result of coupled substitution with $(\text{Cu}, \text{Ag})^+$ for 2Zn^{2+} .
4. The metals in the Maweishan deposit are derived primarily from the proximal quartz diorite, with a small amount derived from the surrounding country rocks.

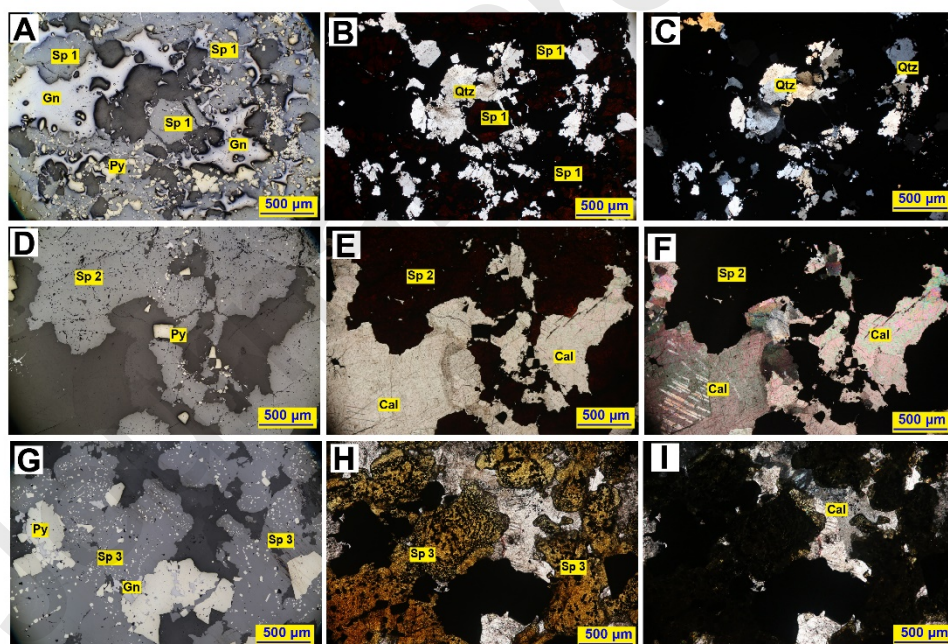
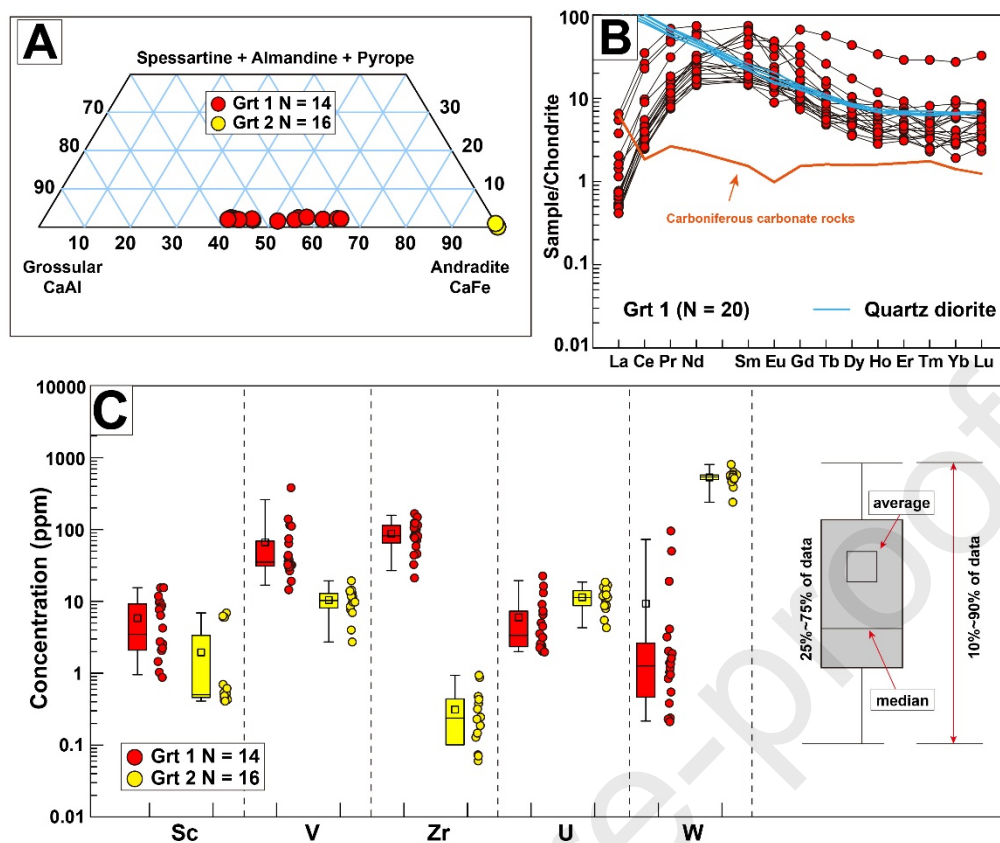


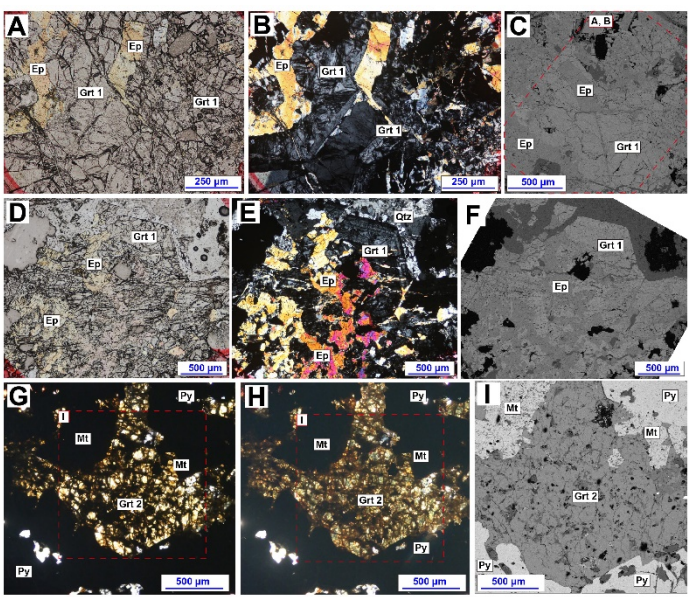
Garnet U–Pb dating of samples from the Maweishan deposit of the Xuancheng ore district within of the Middle and Lower Yangtze River metallogenic belt of eastern China yields a well-defined lower-intercept age of 132.3 ± 1.4 Ma (Fig. A), consistent with a zircon U–Pb age of a proximal quartz diorite intrusion (132.4 ± 0.9 Ma). This strongly suggests that the quartz diorite is genetically associated with the formation of the Maweishan deposit rather than the deposit being a distal part of the Qiaomaishan skarn system. The flat LA-ICP-MS signal spectra for spherulite within the system suggests that the critical metals Cd, In, and Ga were incorporated into the spherulite structure through isomorphous substitution or are present as homogeneously distributed nanoparticles (Fig. B). The strong correlation between Ga and Cu (Fig. C) suggests that Ga was incorporated by coupled substitution in the form $\text{Cu}^+ + \text{Ga}^{3+}$ replacing 2Zn^{2+} . The narrow range of $\delta^{34}\text{S}$ values for sulfides within the system (Fig. D), all of which are slightly heavy compared to the range expected for magmatic-derived fluids (-5 to $+5\%$), suggests that the metals within the deposit and the fluids that formed the deposit were predominantly derived from a magmatic source, namely the proximal quartz diorite, but with minor amounts also derived from the surrounding country rock.











Minerals	Prograde skarn	Retrograde skarn	Quartz–Sulfide stage	Calcite–Sulfide stage
Garnet	—————			
Diopside	—————			
Quartz		—————	—————	
Epidote		—————	—————	
Actinolite		—————	—————	
Serpentine		—————		
Chlorite		—————	—————	
Sericite		—————	—————	
Magnetite		—————	—————	
Pyrite			—————	—————
Chalcopyrite			—————	
Pyrrhotite			—————	
Sphalerite			—————	—————
Galena			—————	—————
Arsenopyrite				—————
Calcite				—————

Note: ————— major
————— minor

Sp 1 Sp 2 Sp 3

